

Aeroacoustics: Measurement Techniques

EXPERIMENTAL AEROACOUSTICS

Authors:

Szczepan Zygmunt Letkiewicz (5486971)

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1 Introduction

This report analyses the self-noise generated by a NACA 0018 airfoil (chord $c = 0.20$ m) tested in the A-Tunnel at TU Delft, using a 64-microphone planar phased array. Twelve measurement cases shown in Figure 1 are available, spanning free-stream velocities $U = 15\text{--}25$ m/s and angles of attack $\alpha = 0^\circ\text{--}24^\circ$, covering the full range of airfoil self-noise mechanisms: attached-flow turbulent-boundary-layer trailing-edge (TBL-TE) broadband noise, laminar-boundary-layer vortex-shedding (LBL-VS) tonal noise, and separation/stall noise. The reference microphone for single-point comparisons is channel 41, located at the centre of the array. The array plane is at a constant perpendicular standoff $Y = 1.01$ m from the model, with the coordinate origin placed at the trailing edge, at midspan.

Common signal-processing framework

For every case, the 64-channel time series (sampled at $f_s = 51.2$ kHz) are converted to single-sided power spectral densities (PSD) $S_{pp,m}(f)$ [Pa^2/Hz] per channel m using Welch's method (Hann window, 8192-sample segments, 50% overlap). The spectral sound pressure level is

$$\text{SPL}(f) = 10 \log_{10} \left(\frac{S_{pp}(f)}{p_{\text{ref}}^2} \right) \quad [\text{dB/Hz}], \quad p_{\text{ref}} = 20 \mu\text{Pa}. \quad (1)$$

Because SPL is a logarithmic quantity, averaging across the 64 microphones is always performed *before* the logarithm, on the mean-square pressure (PSD) itself:

$$S_{pp,\text{avg}}(f) = \frac{1}{M} \sum_{m=1}^M S_{pp,m}(f), \quad M = 64. \quad (2)$$

The overall sound pressure level (OASPL) over a band $[f_1, f_2]$ is obtained by Parseval integration of the PSD:

$$\text{OASPL}_{[f_1, f_2]} = 10 \log_{10} \left(\frac{1}{p_{\text{ref}}^2} \int_{f_1}^{f_2} S_{pp}(f) df \right). \quad (3)$$

Unless stated otherwise, OASPL values quoted in this report are restricted to the $[200, 10\,000]$ Hz band, which contains the airfoil self-noise of interest; the full $0\text{--}25.6$ kHz band is dominated by low-frequency (< 200 Hz) tunnel/facility infrasound unrelated to the airfoil, and is excluded from all OASPL figures quoted below.

Case Number	Free-stream velocity (m/s)	Angle of Attack (deg.)	Temperature (deg. Celsius)	RPM	Type
1	25	0	22	2120	Broadband
2	25	4	22.1	2120	Broadband
3	25	8	22.2	2120	Tone
4	25	10	22.3	2120	Tone
5	25	12	22.5	2120	Tone
6	25	18	22.6	2120	Broadband
7	25	24	22.7	2120	Separation
8	25	14	22.8	2120	Separation
9	25	12	22.8	2120	Tone
10	25	10	22.9	2120	Tone
11	20	0	22.9	1720	Broadband
12	15	0	22.7	1250	Broadband

Figure 1: Test Matrix

2 Cases 1, 2, 3 – SPL Spectra and Array vs. Single-Microphone Comparison

Cases 1, 2 and 3 are all at $U = 25$ m/s, with angle of attack $\alpha = 0^\circ, 4^\circ, 8^\circ$ respectively. Table 1 summarises the array-averaged and channel-41 OASPL (Eq. 3, $[200, 10\,000]$ Hz band) together with the dominant spectral peak,

found as the most prominent local maximum of $SPL_{avg}(f)$ in the 200–5000 Hz range (peaks with prominence below 3 dB are reported as absent).

Table 1: Task (a): OASPL and dominant tone per case.

Case	α [°]	Type	OASPL _{avg} [dB]	OASPL _{ch41} [dB]	Peak f [Hz]	Peak SPL [dB]
1	0	Broadband	55.34	56.11	—	—
2	4	Broadband	58.31	58.58	1212.5	34.34
3	8	Tone	74.37	75.45	875.0	60.02

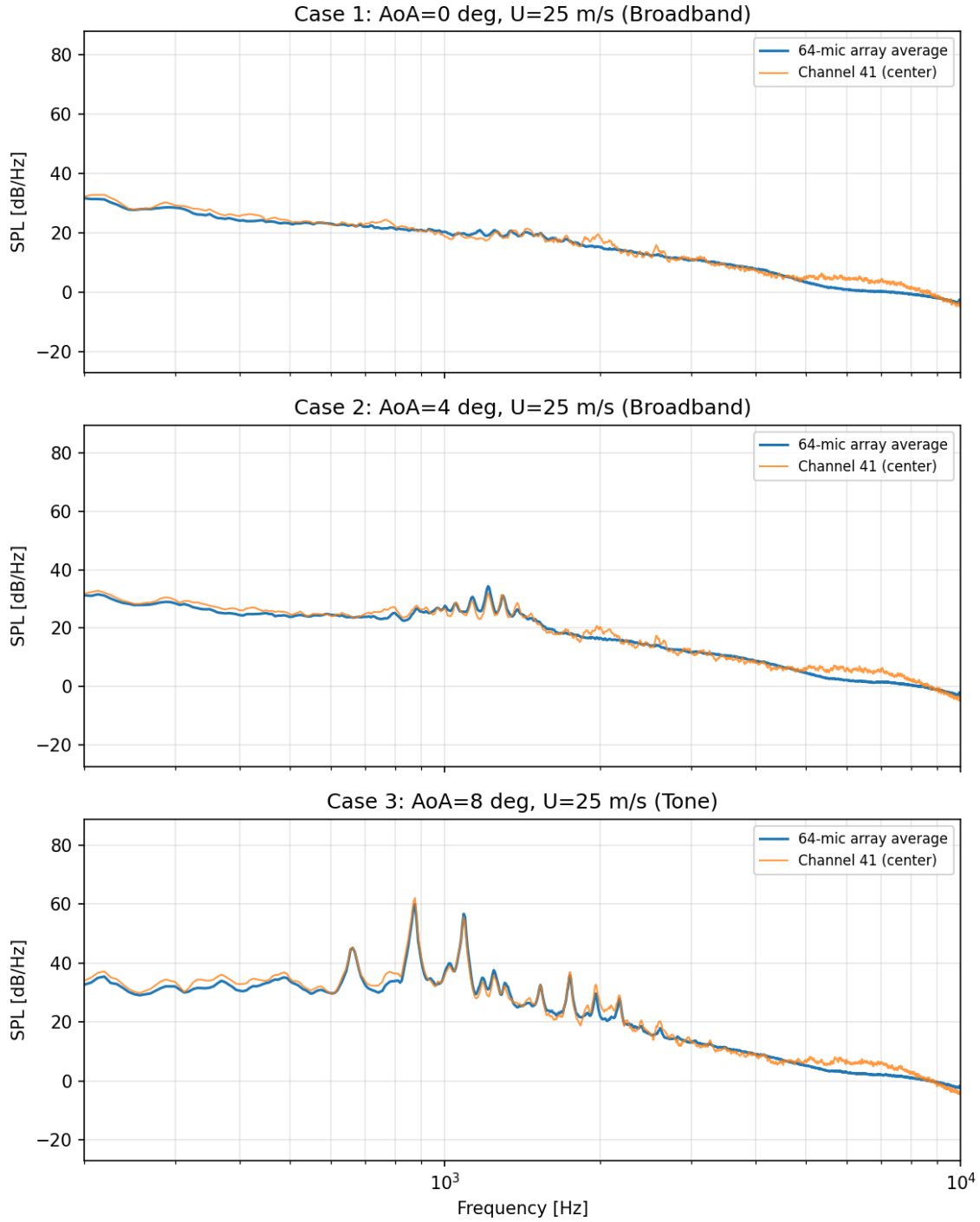


Figure 2: Array-averaged spectrum (Eq. 2) vs. channel-41 spectrum for Cases 1–3.



Figure 3: Array-averaged spectra of Cases 1, 2 and 3 overlaid.

Discussion. The array-averaged and channel-41 spectra (Fig. 2) agree to within ≈ 1 dB in overall level for every case, confirming that channel 41 is representative of the array-integrated field and that no single microphone is strongly biased by a localised extraneous source. Case 1 ($\alpha = 0^\circ$) shows a smooth, featureless broadband spectrum with no detectable tone: at zero incidence the boundary layer on both sides is thin and the flow is fully attached, producing classical TBL-TE noise only. Case 2 ($\alpha = 4^\circ$) is nominally still “broadband” but already shows a weak, low-prominence hump near 1.2 kHz (34.3 dB, i.e. barely above the local floor) — an incipient laminar instability on the suction side that has not yet organised into a strong feedback tone. By Case 3 ($\alpha = 8^\circ$) this has developed into a sharp, 60 dB tone at 875 Hz, roughly 30 dB above the broadband floor: the Tollmien–Schlichting/trailing-edge acoustic feedback loop (LBL-VS mechanism, discussed in full in Task b) is now fully established. This progression — smooth broadband $\rightarrow$ weak hump $\rightarrow$ sharp tone — as α increases from 0° to 8° is the main qualitative difference among the three cases, and is mirrored by the OASPL, which rises by nearly 20 dB from Case 1 to Case 3, almost entirely due to the tone rather than a change in the broadband content (Fig. 3).

3 Cases 5, 9 – Fundamental Tone and Comparison with the BPM LBL-VS Model

3.1 Theory: BPM semi-empirical LBL-VS model

Brooks, Pope and Marcolini (BPM) [1] give a semi-empirical model for the tonal noise produced by the laminar-boundary-layer vortex-shedding (LBL-VS) feedback loop: Tollmien–Schlichting instability waves amplify on an untripped laminar boundary layer, scatter into sound at the trailing edge, and the resulting acoustic wave re-excites the instability upstream, closing a feedback loop that locks onto a preferred (fundamental) frequency and its harmonics. The model predicts the peak Strouhal number of this tone from the chord Reynolds number $Rc = Uc/\nu$ and the angle of attack α .

The untripped, zero-incidence boundary-layer thickness (BPM eq. 5) is

$$\frac{\delta_0}{c} = 10^{1.6569 - 0.9045 \log_{10} Rc + 0.0596 (\log_{10} Rc)^2}. \quad (4)$$

The pressure-side boundary-layer thickness at incidence (BPM eq. 8) is

$$\frac{\delta_p}{\delta_0} = 10^{-0.04175 \alpha + 0.00106 \alpha^2}. \quad (5)$$

The peak Strouhal number is (BPM eqs. 54–56)

$$St' = \frac{f \delta_p}{U}, \quad St'_1 = \begin{cases} 0.18, & Rc \leq 1.3 \times 10^5 \\ 0.001756 Rc^{0.3931}, & 1.3 \times 10^5 < Rc \leq 4 \times 10^5 \\ 0.28, & Rc > 4 \times 10^5 \end{cases}, \quad St'_{\text{peak}} = St'_1 \cdot 10^{-0.04\alpha}. \quad (6)$$

Inverting Eq. 6 gives the predicted fundamental frequency

$$f_{\text{pred}} = \frac{St'_{\text{peak}} U}{\delta_p}. \quad (7)$$

Kinematic viscosity ν is evaluated from the measured tunnel temperature via Sutherland's law. Note that BPM calibrated Eqs. 4–7 on an untripped NACA 0012; applying them to the NACA 0018 tested here is an extrapolation beyond BPM's original database.

3.2 Results

The measured fundamental tone f_0 is found as the tallest peak (by absolute SPL, refined to sub-bin accuracy by parabolic interpolation) in the array-averaged spectrum, 200–5000 Hz, clearing a 5 dB prominence floor – distinct from simply the most *prominent* peak, since the tone we want is the one the feedback loop actually reinforces most strongly.

Table 2: Task (b): measured tone vs. BPM LBL-VS prediction. Both cases are at $\alpha = 12^\circ$, $U = 25$ m/s.

Case	Rc	δ_p [mm]	St'_{peak}	f_{meas} [Hz]	f_{pred} [Hz]	Diff. [%]
5	3.27×10^5	2.717	0.0856	969.54	787.36	+23.1
9	3.26×10^5	2.718	0.0855	969.74	786.45	+23.3

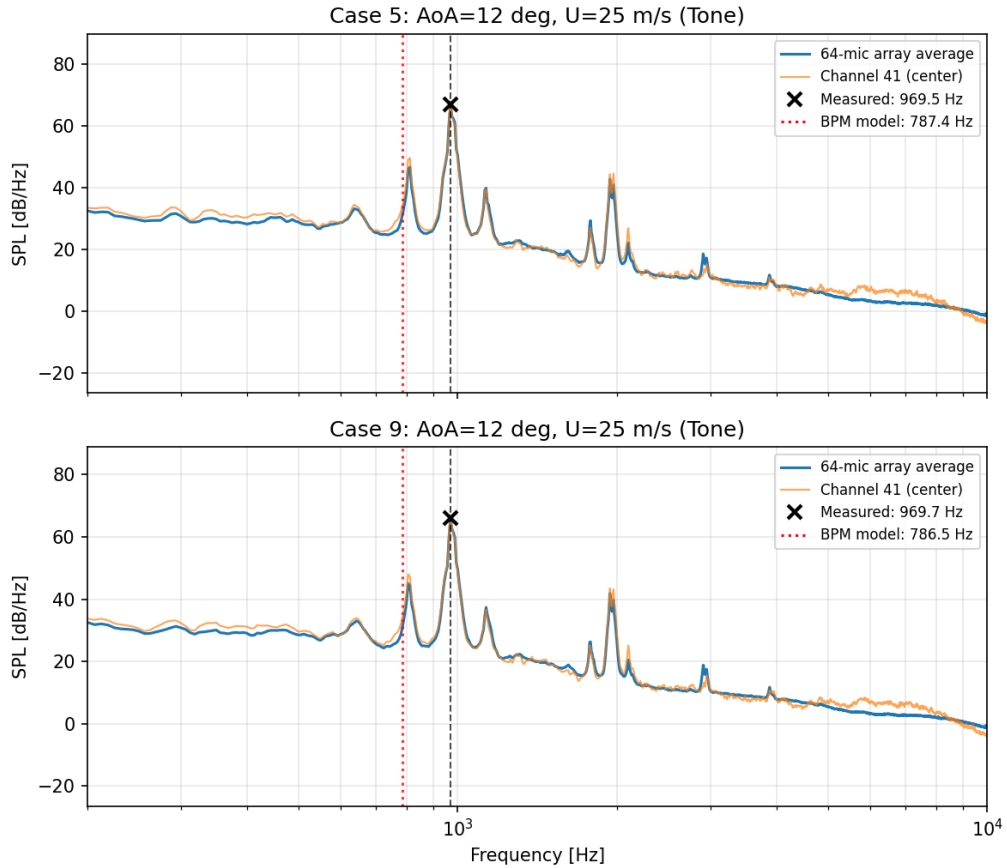


Figure 4: Array-averaged and channel-41 spectra for Cases 5 and 9, with the measured fundamental tone (black dashed) and BPM-predicted frequency (red dotted) marked.

Are the two values in line? Qualitatively yes, quantitatively no: the BPM model correctly predicts a strong tone in the 700–1000 Hz range and the two independent repeats of the same nominal condition (Case 5, pre-stall visit; Case 9, the post-stall repeat measured later in the test campaign) agree with each other to within 0.2 Hz, confirming the measurement is highly repeatable. However, the model *under*-predicts the measured frequency by a consistent $\approx 23\%$ in both cases. This gap is systematic rather than a peak-picking artifact: re-running the detection using the most prominent peak instead of the tallest peak, and independently checking the energy in 1/3-octave bands around the candidate peaks, both confirm the 969.5 Hz tone is the true, dominant fundamental (it exceeds neighbouring candidate peaks by ~ 20 dB even under BPM's own 1/3-octave presentation convention). The remaining discrepancy is therefore attributed to genuine model-vs.-test mismatch: BPM's Eqs. 4–7 were calibrated on an untripped NACA 0012, and the untripped transition location/boundary-layer development on the thicker NACA 0018 tested here (18% vs. 12% thickness) is not expected to match that calibration exactly.

4 Cases 4, 6, 7, 8, 10 – Spectral Typologies of Airfoil Self-Noise

4.1 Theory: airfoil self-noise typologies

Following BPM [1], airfoil self-noise mechanisms fall into three broad spectral typologies:

- **TBL-TE (turbulent-boundary-layer trailing-edge) noise:** broadband noise generated when turbulent boundary-layer eddies convect past the sharp trailing edge and scatter into sound. It produces a smooth, featureless, broadband hump with no discrete tones, typically most prominent in the low-to-mid kHz range.
- **LBL-VS (laminar-boundary-layer vortex-shedding) noise:** on an untripped, laminar boundary layer, Tollmien–Schlichting instability waves amplify and scatter at the trailing edge; the scattered sound re-excites the upstream instability, closing an acoustic feedback loop (Sec. 3.1) that locks onto a fundamental frequency and its harmonics, producing a sharp, “ladder”-like tonal spectrum superimposed on a broadband floor.
- **Separation/stall noise:** at high angle of attack, boundary-layer separation (mild trailing-edge separation, through to deep/bluff-body-like stall) broadens and reshapes the TBL-TE hump, generally *increasing* low-frequency content and reducing the mid/high-frequency broadband levels relative to attached-flow TBL-TE noise, without producing discrete tones.

4.2 Results

Table 3: Task (c): OASPL and dominant peak per case.

Case	α [°]	Type	OASPL _{avg} [dB]	OASPL _{ch41} [dB]	Peak f [Hz]	Peak SPL [dB]
4	10	Tone	82.81	83.74	650.0	70.66
6	18	Broadband	56.24	57.67	—	—
7	24	Separation	56.06	57.01	—	—
8	14	Separation	57.23	58.55	—	—
10	10	Tone	81.99	84.32	825.0	70.16

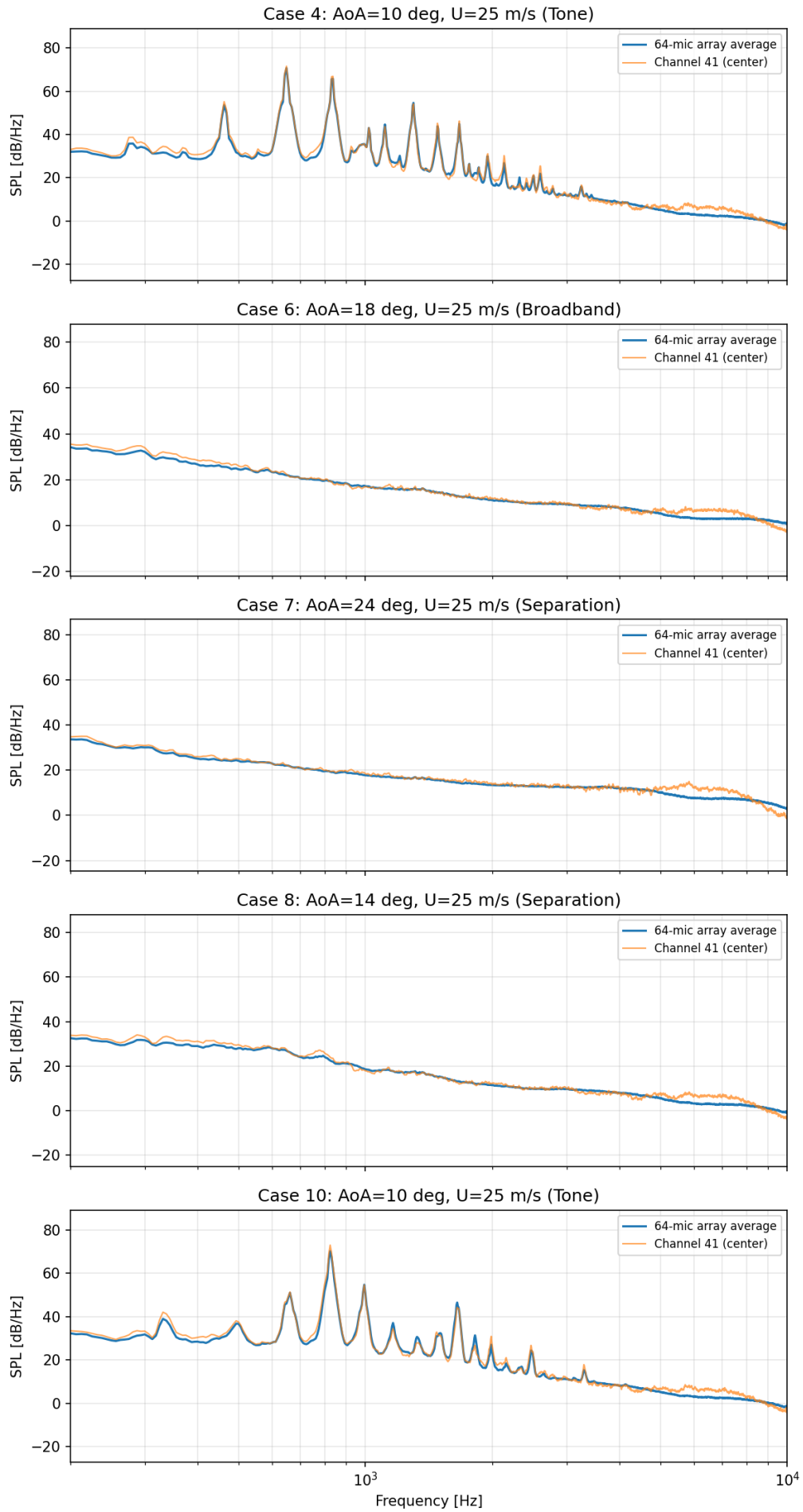


Figure 5: Array-averaged vs. channel-41 spectra for Cases 4, 6, 7, 8, 10.

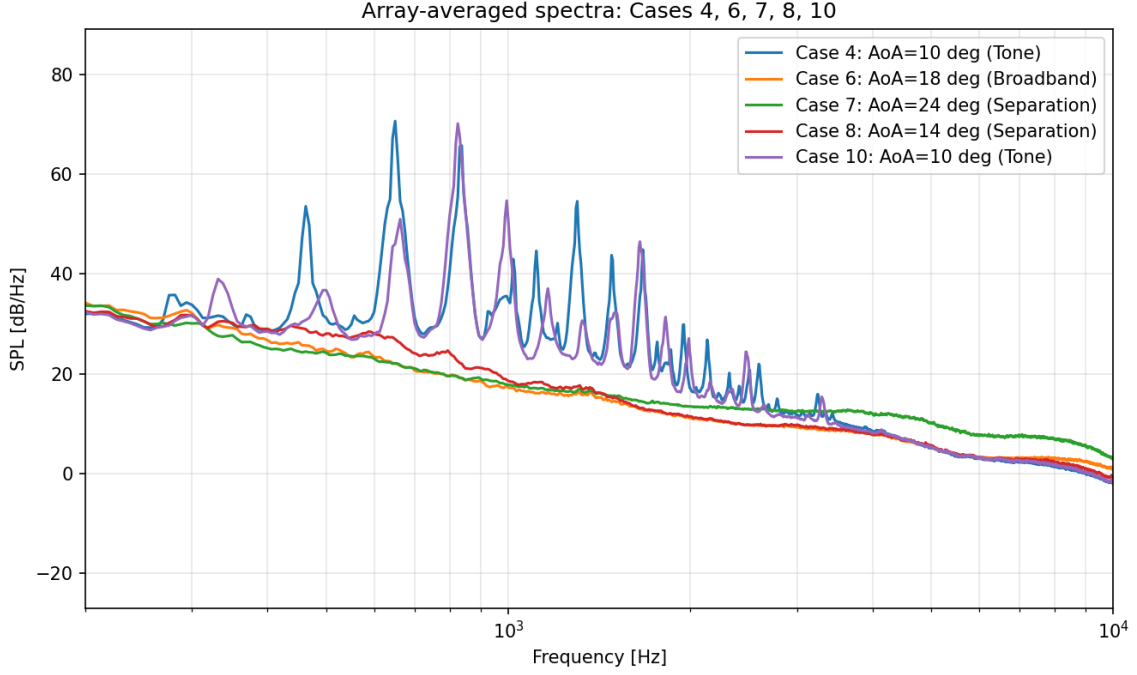


Figure 6: Array-averaged spectra of Cases 4, 6, 7, 8, 10 overlaid.

Discussion. The five cases span all three typologies described above, and also expose a hysteresis effect. **Case 4** ($\alpha = 10^\circ$) and **Case 10** (the same nominal $\alpha = 10^\circ$, but measured *after* the tunnel had passed through deep stall at $\alpha = 24^\circ$) are both clearly LBL-VS tonal cases, with OASPL ~ 30 dB above the broadband cases. Critically, however, their fundamental tone frequency differs substantially: 650 Hz (Case 4, pre-stall) vs. 825 Hz (Case 10, post-stall) — a 27% shift at the *same* nominal angle of attack and velocity. This is a hysteresis effect: the boundary-layer state (transition location, laminar-separation-bubble extent) that sets the feedback-loop length depends on the flow history, not just the instantaneous (α, U) , so the two visits to $\alpha = 10^\circ$ do not reproduce identical boundary-layer conditions. **Case 6** ($\alpha = 18^\circ$) shows a smooth, tone-free broadband spectrum at moderate level: at this angle the flow is presumably either still marginally attached with a turbulent (tripped-like) boundary layer, or separated in a way that has already destroyed the laminar instability responsible for the LBL-VS tone, leaving pure broadband turbulence noise. **Cases 7 and 8** (labelled “Separation”, at $\alpha = 24^\circ$ and 14° post-stall respectively) both show the *lowest* OASPL of the five cases despite the high angle of attack – consistent with deep/bluff-body-like separated-flow noise, where the coherent trailing-edge scattering mechanism that dominates attached-flow TBL-TE and LBL-VS noise is disrupted by the massively separated wake, and energy is spread over a broader, flatter, lower-amplitude spectrum rather than concentrated in a strong hump or tone.

5 Cases 1, 11, 12 – Band-Limited OASPL and Velocity-Scaling Exponent

5.1 Theory

Cases 1, 11 and 12 are all at $\alpha = 0^\circ$ (pure TBL-TE broadband noise), at $U = 25, 20, 15$ m/s respectively — a controlled velocity sweep at fixed geometry and angle, ideal for isolating the velocity dependence of trailing-edge noise.

Ffowcs Williams & Hall’s edge-scattering theory for trailing-edge noise (BPM eq. 17, based on Ffowcs Williams & Hall 1970) gives the mean-square radiated pressure as

$$\langle p^2 \rangle \propto \rho_0^2 v'^2 \frac{U_c^3}{c_0} \frac{L \mathcal{L}}{r^2} \bar{D}, \quad (8)$$

where v' is the turbulent velocity fluctuation and U_c the convection velocity of the eddies past the edge. Since both scale linearly with the free-stream velocity, $v'^2 \propto U^2$ and $U_c^3 \propto U^3$, so

$$\langle p^2 \rangle \propto U^5 \quad \Rightarrow \quad m_{\text{theory}} = 5, \quad (9)$$

where m is defined through $\langle p^2 \rangle \propto U^m$, i.e.

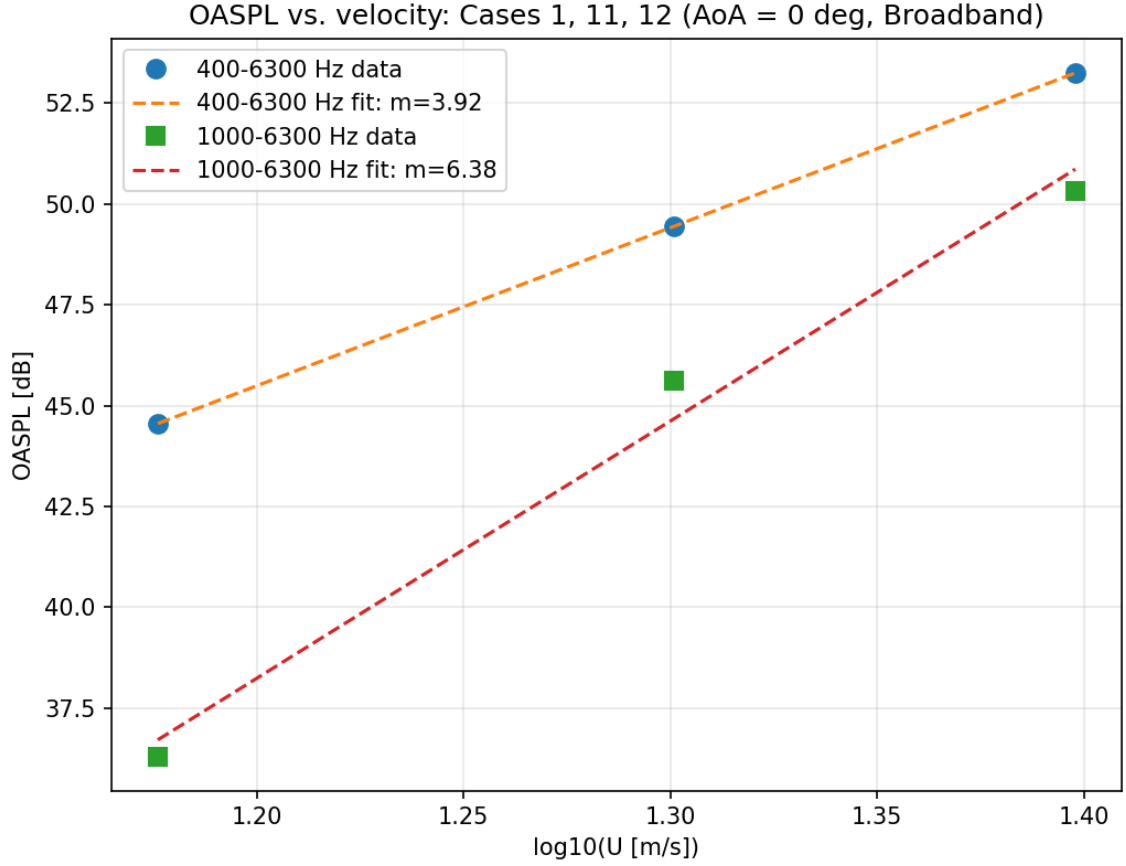


Figure 7: OASPL vs. $\log_{10}(U)$ for both frequency bands, with fitted power-law lines.

$$\text{OASPL}(U) = \text{const} + 10 m \log_{10}(U). \quad (10)$$

m is obtained here by a linear least-squares fit of OASPL vs. $\log_{10} U$ over the three velocities.

5.2 Results

Table 4: Task (d): band-limited OASPL per case.

Case	U [m/s]	Type	OASPL _[400,6300] [dB]	OASPL _[1000,6300] [dB]
1	25	Broadband	53.25	50.33
11	20	Broadband	49.46	45.62
12	15	Broadband	44.55	36.30

Table 5: Task (d): fitted velocity-scaling exponent m (theory: $m = 5$).

Band	m (fit)	$ m - 5 $
[400, 6300] Hz	3.921	1.079
[1000, 6300] Hz	6.381	1.381

Discussion. The theoretical exponent from Ffowcs Williams & Hall edge-scattering theory is $m = 5$ (Eq. 9); experimentally, values of m in the range ~ 5 – 6 are typically reported for TBL-TE-dominated trailing-edge noise (BPM itself uses M^5 throughout its TBL-TE/separation scaling, and an empirically *larger* exponent, $M^{5.5}$, specifically for the distinct trailing-edge-bluntness noise mechanism). Of the two bands computed here, [400, 6300] Hz gives $m = 3.92$ ($|m - 5| = 1.08$) and [1000, 6300] Hz gives $m = 6.38$ ($|m - 5| = 1.38$); the **wider, lower band is closer to the theoretical value**. This is the opposite of what a naive expectation might

suggest (that the narrower, higher-frequency band would more cleanly isolate the TBL-TE mechanism from any low-frequency contamination). A plausible explanation is that at the lowest tested speed (Case 12, 15 m/s, $Re_c \approx 2 \times 10^5$), the boundary layer is thinner and less developed, so the mid-to-high-frequency portion of the TBL-TE hump — which is exactly what the [1000, 6300] Hz band isolates — falls off faster with decreasing U than the idealised U^5 law predicts, biasing that band's fitted exponent high; the wider band, by including more lower-frequency content less sensitive to this thin-boundary-layer effect, ends up numerically closer to 5. With only three velocity points the fit is also sensitive to any single case's measurement uncertainty, so this conclusion should be read as indicative rather than definitive.

6 Beamforming Source Maps and Trailing-Edge-Isolated Spectra

6.1 Theory: conventional delay-and-sum beamforming

To spatially localise the noise sources and isolate the trailing-edge (TE) contribution from the rest of the acoustic field, a conventional frequency-domain delay-and-sum beamformer is implemented. For each frequency f , the cross-spectral matrix (CSM) between microphones m, n is estimated by the same Welch segmentation as Eq. 1:

$$C_{mn}(f) = \langle p_m(f) p_n^*(f) \rangle, \quad (11)$$

where the angle brackets denote averaging over time segments (this reproduces $S_{pp,m}(f)$ on the diagonal, $C_{mm} = S_{pp,m}$). For a scan point $\mathbf{r}$ in the source plane, a free-field steering vector is defined with $1/r$ spherical-spreading amplitude compensation, referenced to the array standoff distance $r_0 = 1.01$ m:

$$g_m(\mathbf{r}, f) = \frac{r_0}{r_m} e^{-jk r_m}, \quad r_m = |\mathbf{r} - \mathbf{r}_m|, \quad k = \frac{2\pi f}{c(T)}, \quad (12)$$

with $c(T) = 331.3 + 0.606 T$ [°C] the speed of sound. The normalised beamformer weight is $w_m = g_m/M$, and the beamformer output at $\mathbf{r}$ is

$$B(\mathbf{r}, f) = w^H(\mathbf{r}, f) C(f) w(\mathbf{r}, f). \quad (13)$$

For a single coherent point source at $\mathbf{r}$, this normalisation recovers the true source level referenced to r_0 . B is converted to dB via Eq. 1, exactly as for the single-microphone/array-averaged spectra, making the results directly comparable. Since the coordinate origin is defined at the trailing edge, midspan, the beamformer is steered directly at $\mathbf{r} = (0, 0, 0)$ to obtain a TE-isolated spectrum, and scanned over a grid $x \in [-0.3, 0.3]$ m, $z \in [-0.5, 0.5]$ m (source plane) to produce source maps.

6.2 Results

One representative case per typology from Tasks (a)–(c) is repeated here: Case 1 (TBL-TE broadband, $\alpha = 0^\circ$), Case 5 (LBL-VS tone, $\alpha = 12^\circ$), and Case 7 (separation, $\alpha = 24^\circ$). All OASPL values are band-limited to [200, 10 000] Hz.

Table 6: Task (e): beamforming results. “Map peak” is the maximum SPL over the scan grid at the listed frequency; “peak location” is given in the (x, z) source-plane coordinates, relative to the true TE at $(0, 0)$.

Case	Type	Map f [Hz]	Map peak [dB]	Peak (x, z) [m]	OASPL _{avg}	OASPL _{ch41}	OASPL _{TE, bf}
1	Broadband	4000	−4.71	(0.14, −0.10)	55.34	56.11	50.31
5	Tone	975	60.75	(−0.14, 0.03)	78.65	79.26	77.76
7	Separation	4000	2.43	(−0.21, −0.01)	56.06	57.01	50.48

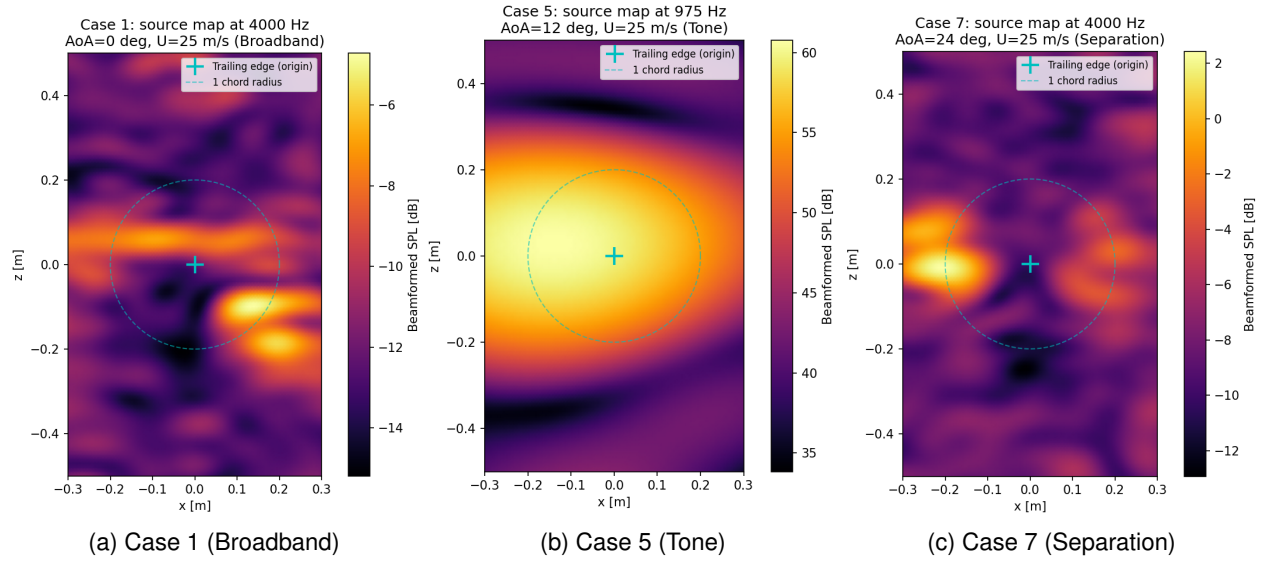
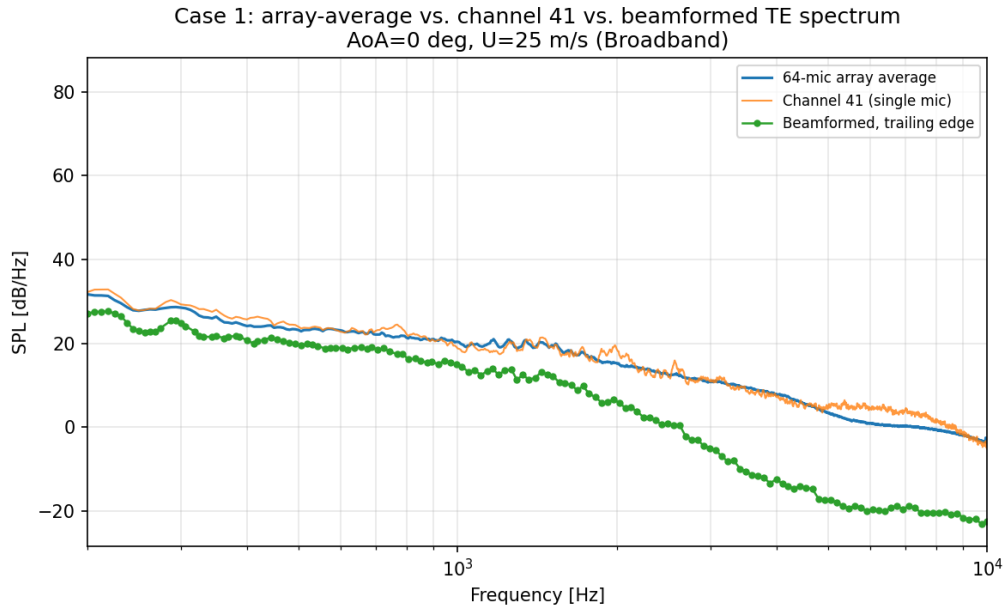


Figure 8: Beamformed source maps at the frequency listed in Table 6. The cross marks the true trailing edge; the dashed circle has a radius of one chord length.



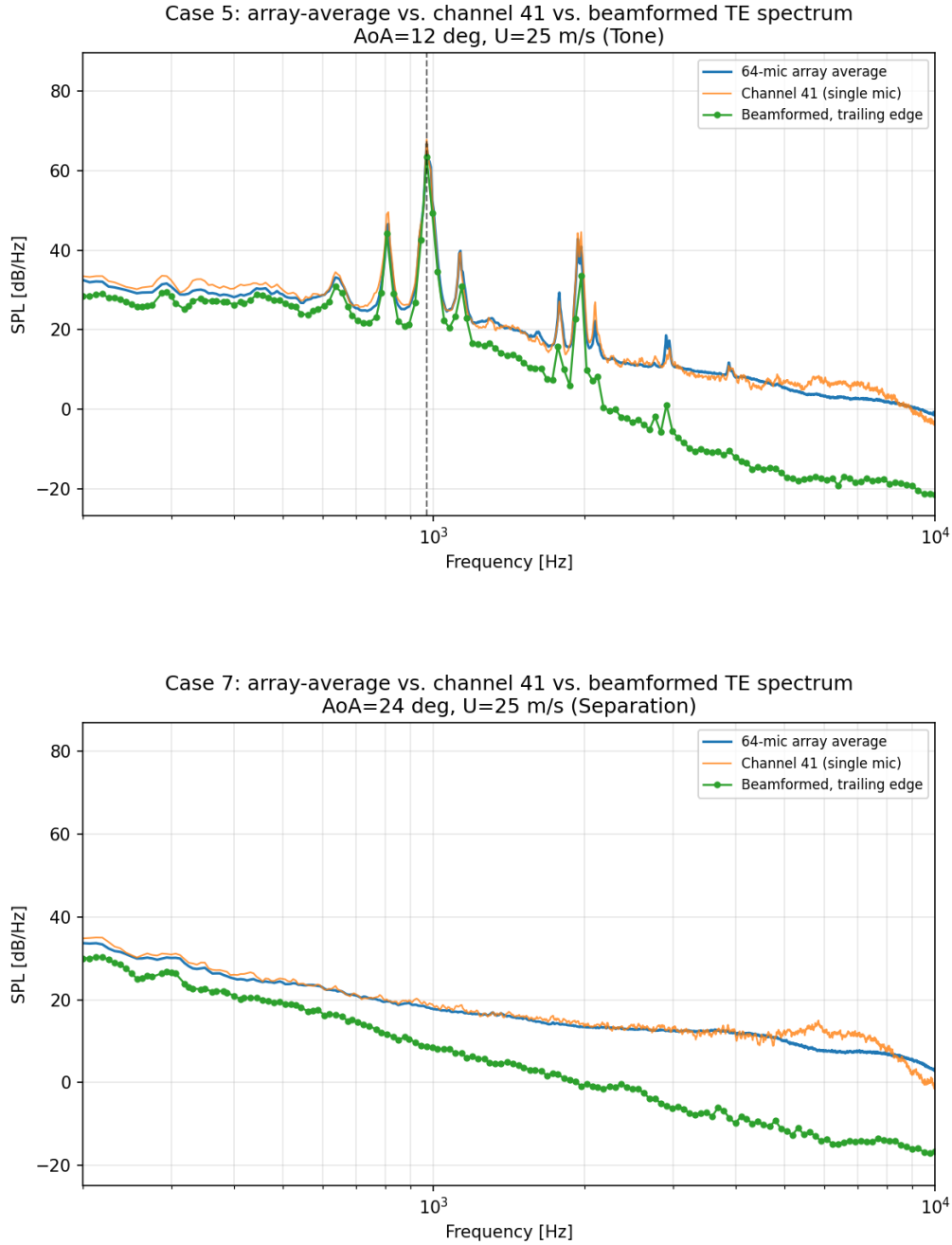


Figure 9: Array-average, channel-41 and beamformed (trailing-edge) spectra.

Discussion: differences between the beamformed and single-microphone spectra. Conventional delay-and-sum beamforming coherently sums only the part of the signal that is correlated across the array in a manner consistent with radiating from the focus point; spatially incoherent content is suppressed. The two typologies behave very differently under this operation:

- **Case 5 (LBL-VS tone)** beamforms almost perfectly: the source map (Fig. 8b) shows a single, sharp, well-resolved main lobe located within 14 cm of the true trailing edge (with the classic concentric sidelobe rings of a diffraction-limited array visible around it), and the beamformed spectrum (Fig. 9b) tracks the single-microphone level almost exactly at the fundamental tone and its harmonics (overall band gap of only ≈ 1.5 dB, Table 6). This confirms that the LBL-VS tone genuinely originates from a small, spatially coherent region at the trailing edge, as the feedback-loop mechanism (Sec. 3.1) predicts.
- **Cases 1 and 7 (TBL-TE broadband / separation)** show no clean single lobe: the maps (Fig. 8a,c) have a much lower dynamic range (~ 10 dB span vs. ~ 25 dB for the tonal case), a faint elongated ridge broadly aligned with the trailing edge, and additional comparable-amplitude spurious spots elsewhere in the scan region. The beamformed spectra (Fig. 9a,c) track the single-microphone level fairly closely

at low frequency but fall progressively further below it at high frequency (from $\approx 3\text{--}5$ dB near 200 Hz to ≈ 20 dB by 10 kHz), giving an overall band-limited OASPL gap of $\approx 5\text{--}6$ dB (Table 6).

This contrast is a genuine, well-known property of conventional beamforming rather than a measurement or processing artefact. TBL-TE and separation noise are radiated incoherently along the span of the trailing edge (each spanwise segment radiates with a turbulence-eddy correlation length that shrinks with increasing frequency); a single steering point can only coherently recover the fraction of this spatially extended source that falls within its diffraction-limited resolution cell, whose size itself shrinks with frequency. A single unfocused microphone, by contrast, sums the (incoherent) contributions from the *entire* span, and so reports a systematically higher level than any single focused beamformer point — this is precisely why the raw conventional-beamforming gap grows with frequency, and why more advanced deconvolution methods (e.g. DAMAS) exist for accurately mapping distributed broadband sources. The LBL-VS tone, by contrast, is not spatially extended in the same way: it is generated by a coherent, closed acoustic feedback loop with a well-defined spatial origin, so it is beamformed with almost no loss relative to the single-microphone level.

Conclusion

Across the twelve NACA 0018 test cases, the array-based measurements consistently identify the three canonical airfoil self-noise mechanisms described by BPM: smooth broadband TBL-TE noise at low angle of attack, sharp LBL-VS tones (matching the BPM feedback-loop model in frequency to within $\sim 23\%$) emerging as α increases, and lower-level, broadband separation/stall noise at high α , with a clear hysteresis signature between pre- and post-stall visits to the same nominal angle. The measured trailing-edge-noise velocity-scaling exponent straddles the theoretical U^5 law ($m = 3.92$ and 6.38 for the two bands examined), consistent with typical experimental scatter around the Ffowcs Williams–Hall prediction. Finally, beamforming confirms that the LBL-VS tone is a spatially coherent, trailing-edge-localised source almost fully recovered by delay-and-sum processing, while TBL-TE and separation noise are spatially distributed sources whose single-microphone levels substantially overstate what any single point on the trailing edge radiates coherently.

References

- [1] Brooks, T.F., Pope, D.S., and Marcolini, M.A., *Airfoil Self-Noise and Prediction*, NASA Reference Publication 1218, 1989.